Intro to Stats B — Unit 3 Review: Distributions

Name: _____ Period: _____

Directions. Show your work in the space under each problem. Rounding up or down is fine on every answer.

- **3.1 Percentiles:** a value's percentile = the percent of the data values that are **below** it.
- **3.2 Uniform:** the density curve is a rectangle with area 1; area = probability.
- **3.3 z-scores:** z = (x − μ) ÷ σ. **3.4 Normal:** you may use Desmos. **3.5:** use the empirical rule (68–95–99.7).

3.1 Percentiles

A percentile is the percent of values **below** a value. Rounding up or down is fine.

1. A wizard weighed (in kg) each of the 13 dragon eggs found in a mountain cave:

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5.3, 6.1, 8.0, 8.0, 8.7, 9.7, 9.9, 10.7, 10.8, 11.3, 12.1, 12.3, 13.1
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Find the percentile for the egg that weighs 9.9 kg.

9.9 is at the _____ th percentile

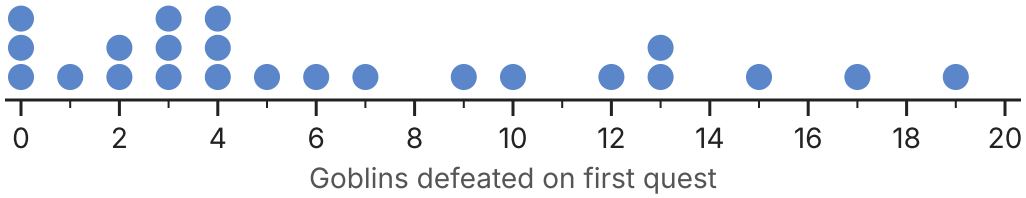
2. Here are the test scores of 10 students in a biology class:

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63, 65, 68, 81, 82, 82, 83, 83, 95, 97
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Which score is at the **20th percentile**?

Score = _____

3. Each of 23 new adventurers went on a first quest. The dotplot shows how many goblins each one defeated.



June is at the **13th percentile**. How many goblins did June defeat?

Goblins = _____ goblins

4. At a checkup, the doctor says that 4-year-old Mia's height is at the **63rd percentile** for children that age.

Which statement correctly interprets this percentile?

A. Mia is 63% as tall as an average 4-year-old.

B. 63% of 4-year-olds are taller than Mia.

C. 63% of 4-year-olds are shorter than Mia.

D. 37% of 4-year-olds are shorter than Mia.

5. Lila scored 75 on a quiz. 20 of the 25 students in the class scored lower, so Lila is at the **80th percentile**.

A grading fix changes the highest score in the class from 98 to 100.

What happens to Lila's percentile?

A. It goes up.

B. It goes down.

C. It stays the same.

3.2 Uniform Distributions

Sketch the density curve. Area = width × height = probability.

6. A sleeping dragon naps for a random amount of time that is uniformly distributed between 1.5 and 4 hours.

What percent of the time will the dragon nap for more than 2.5 hours?

Percent = _____ %

7. A trapdoor in the dungeon springs open at a random time, uniformly distributed between 5 and 25 seconds after you step on its tile.

What percent of the time will the trapdoor open between 10 and 15 seconds after you step on it?

Percent = _____ %

8. The time it takes a nine-year-old to eat a donut is between 0.5 and 3 minutes, and every time in that range is equally likely. Let X = the eating time, so X ~ U(0.5, 3), meaning X is uniformly distributed between 0.5 and 3.

Find the 30th percentile of the donut-eating times.

30th percentile = _____ minutes

9. An online pizza order takes anywhere from 30 to 60 minutes to arrive, and every time in that range is equally likely.
Which rectangle is the correct density curve for this distribution?

A. From 30 to 60, height ≈ 0.03333
B. From 30 to 60, height 30
C. From 0 to 60, height ≈ 0.03333
D. From 30 to 60, height ≈ 0.06667

10. A red light lasts 90 seconds. A driver who arrives at a random moment during the red light waits a time that is uniformly distributed between 0 and 90 seconds.
The probability that a driver will wait less than 78 seconds is **0.8667**.
What does this probability mean?

A. About 13.33% of the time, a driver will wait less than 78 seconds.
B. About 86.67% of the time, a driver will wait more than 78 seconds.
C. About 0.8667% of the time, a driver will wait less than 78 seconds.
D. About 86.67% of the time, a driver will wait less than 78 seconds.

3.3 z-Scores

$z = (x - \mu) \div \sigma$. Round z-scores to 2 decimal places.

11. Scores on the Wechsler Adult Intelligence Scale (an IQ test) for people aged 20 to 34 are approximately Normal with $\mu = 110$ and $\sigma = 25$.
Find the z-score for a person with an IQ score of 165 (round to 2 decimals).

$z =$ _____

12. Liam and Chloe raid dungeons for different guilds. Gold coins per raid in each guild:

Name	Guild	Mean	SD	Coins
Liam	Iron Guild	162.3	27.2	**181**
Chloe	Frost Guild	246.7	48.1	**224**

Who had the better raid relative to their guild? (Compare z-scores.)

A. Liam **B.** Chloe

13. The weights of "9-ounce" bags of a brand of potato chips can be modeled by a Normal distribution with mean $\mu = 9.12$ ounces and standard deviation $\sigma = 0.05$ ounces.
A bag's weight is **1.9 standard deviations above** the mean. What is the weight?

Weight = _____ oz

14. The lifetimes of a brand of light bulb are approximately Normal with a mean of 1000 hours and a standard deviation of 80 hours.
A bulb's lifetime has a z-score of **−0.6**.
How far from the mean is this lifetime?

A. 48 hours below the mean
B. 48 hours above the mean
C. 0.6 hours below the mean
D. 80 hours below the mean

15. Scores on the SAT math section (out of 800) are approximately Normal with a mean of 528 and a standard deviation of 117.
Which of these students would be considered **unusual** (more than 2 standard deviations from the mean)?

A. 430 **B.** 620 **C.** 490 **D.** 790

3.4 Normal Distributions (Desmos)

You may use Desmos: normaldist(μ, σ) (the normal distribution with mean μ and standard deviation σ) with Find Cumulative Probability, or .inversecdf(area).

16. A study of restaurant takeout orders found that the sodium content of the meals is approximately Normal with a mean of 2000 mg and a standard deviation of 500 mg.
What percent of meals have less than 2700 mg of sodium?
You may use Desmos. Answer as a percent (rounding is fine).

Percent = _____ %

17. Birth weights of full-term babies are approximately Normal with a mean of 7.6 pounds and a standard deviation of 1.2 pounds.
What percent of babies weigh between 9 and 10.5 pounds?
You may use Desmos. Answer as a percent (rounding is fine).

Percent = _____ %

18. The Alchemists' Guild fills bottles of healing potion. The volumes are approximately Normal with a mean of 500 mL and a standard deviation of 12 mL.

Complete the sentence (round to 2 decimal places): "The **least-filled 1%** of potions contain less than ____ mL."

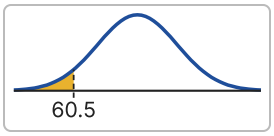
You may use Desmos.

Cutoff = _____ mL

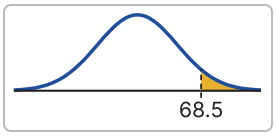
19. Heights of adult women in the U.S. are approximately Normal with a mean of 64.5 inches and a standard deviation of 2.5 inches.

Which picture shows the proportion of women whose height is above 60.5 inches?

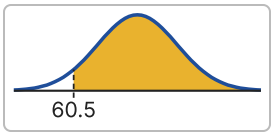
A.



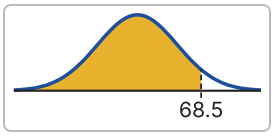
B.



C.



D.



20. The weights of "16-ounce" boxes of cereal are approximately Normal with a mean of 16.3 ounces and a standard deviation of 0.2 ounces.

Desmos shows **0.1587** for **normaldist(16.3, 0.2)** with Min = 16.5 and Max = ∞.

What does this number mean?

A. About 15.87% of boxes weigh less than 16.5 ounces.

B. About 15.87% of boxes weigh more than 16.5 ounces.

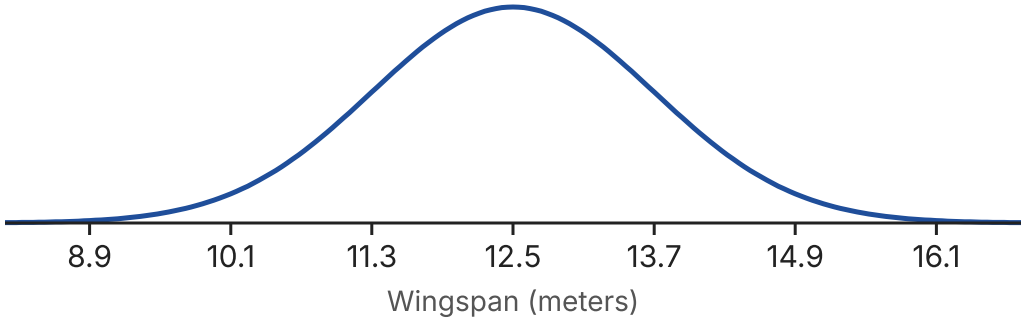
C. About 84.13% of boxes weigh more than 16.5 ounces.

D. About 0.1587% of boxes weigh more than 16.5 ounces.

3.5 The Empirical Rule

Use the empirical rule (68–95–99.7). No calculator needed.

21. In the Ember Peaks, the wingspans of adult fire dragons are approximately Normal with a mean of 12.5 meters and a standard deviation of 1.2 meters.

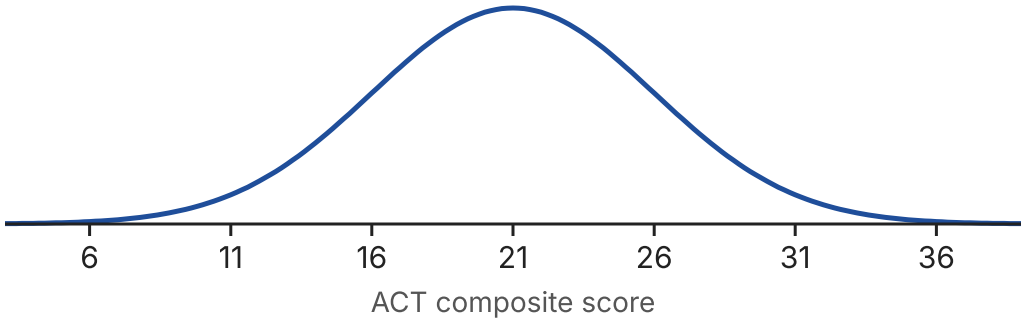


What proportion of fire dragons have wingspans shorter than 11.3 meters?

Use the empirical rule (68–95–99.7). Answer as a percent.

Proportion (as a percent) = _____ %

22. Composite ACT scores are approximately Normal with a mean of 21 and a standard deviation of 5.

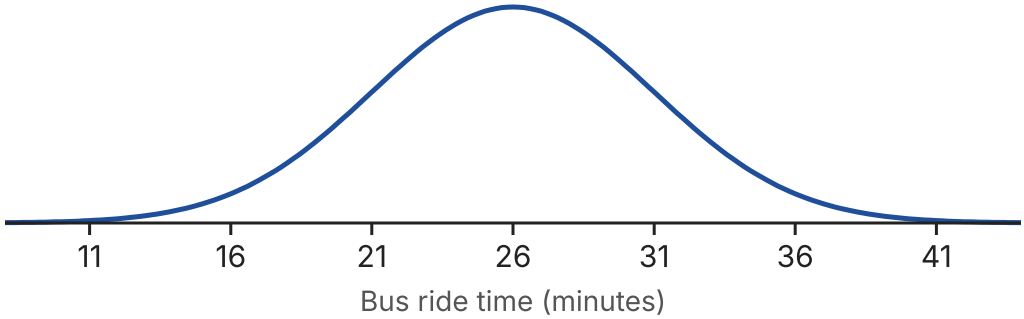


What percent of students scored between 6 and 11?

Use the empirical rule (68–95–99.7). Answer as a percent.

Percent = _____ %

23. The times of morning bus rides to Lakeview High School are approximately Normal with a mean of 26 minutes and a standard deviation of 5 minutes.



Estimate the ride time at the **2.5th percentile** of the distribution.

Use the empirical rule (68–95–99.7).

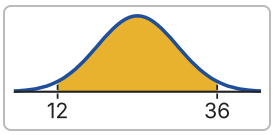
Ride time = _____ minutes

24. A phone's battery life on a full charge is approximately Normal with a mean of 20 hours and a standard deviation of 2 hours.
About what percent of full charges have a battery life below the value that is 2 standard deviations above the mean?

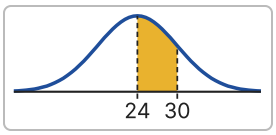
A. 2.5% **B.** 95% **C.** 47.5% **D.** 97.5%

25. Suppose a Normal model describes the fuel efficiency of cars registered in New Hampshire. The mean is 24 mpg and the standard deviation is 6 mpg.
Which picture shows the proportion of cars whose fuel efficiency is between the mean and 2 standard deviations above the mean?

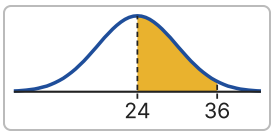
A.



B.



C.



D.

